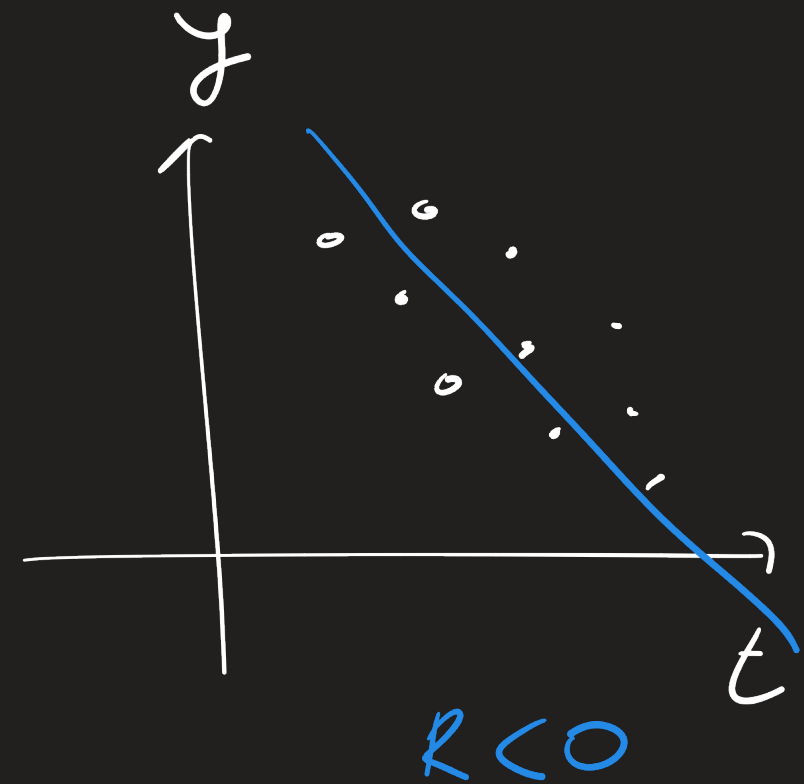
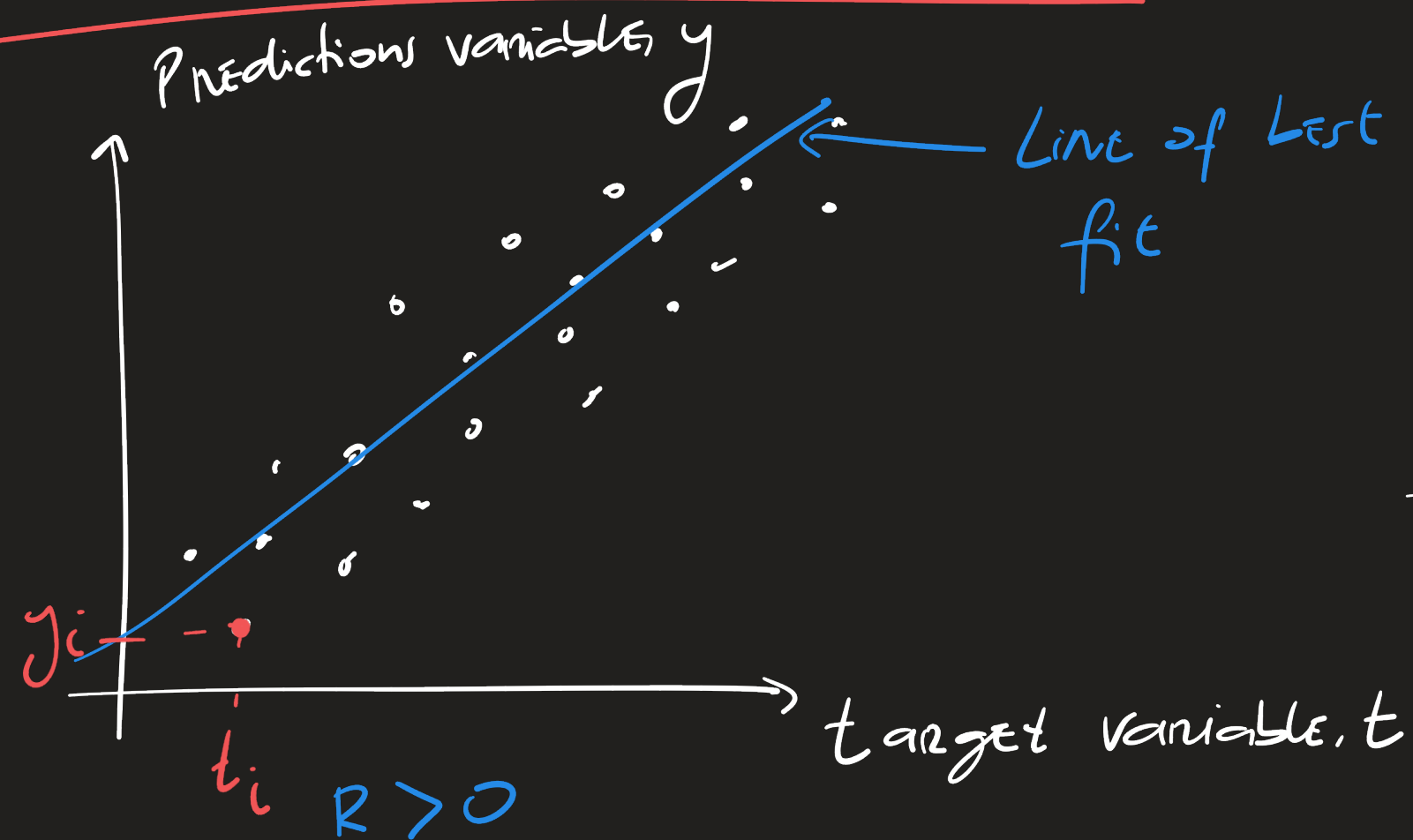


Pearson's Correlation Coefficient $R \in [-1, 1]$



Coefficient of Determination, $R^2 \in [0, 1]$

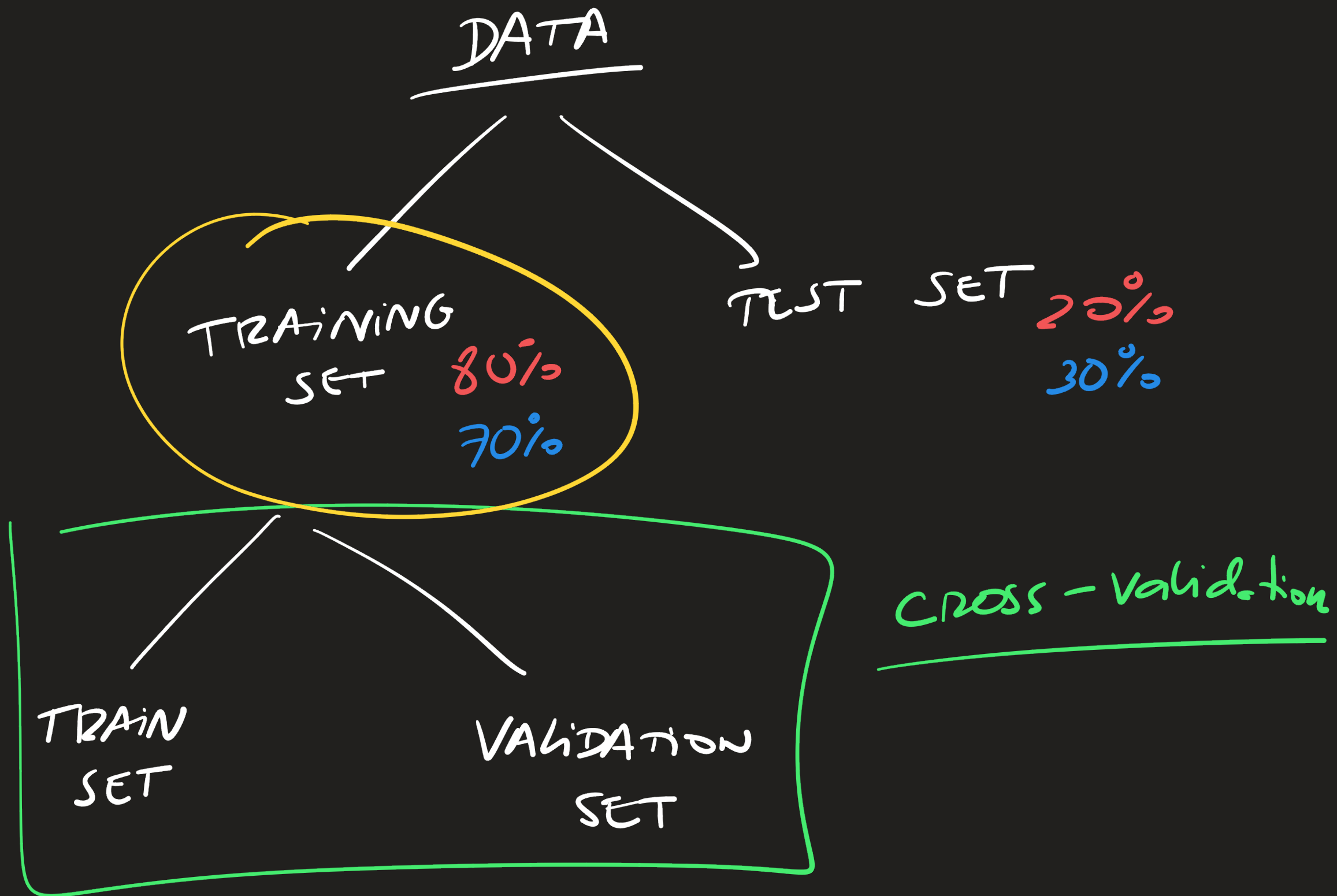
$$R = \frac{\text{COV}(t, y)}{\sigma_t \cdot \sigma_y}$$

Quantile - Quantile (Q-Q) Plot

Consider t and y as two
Random Variables.

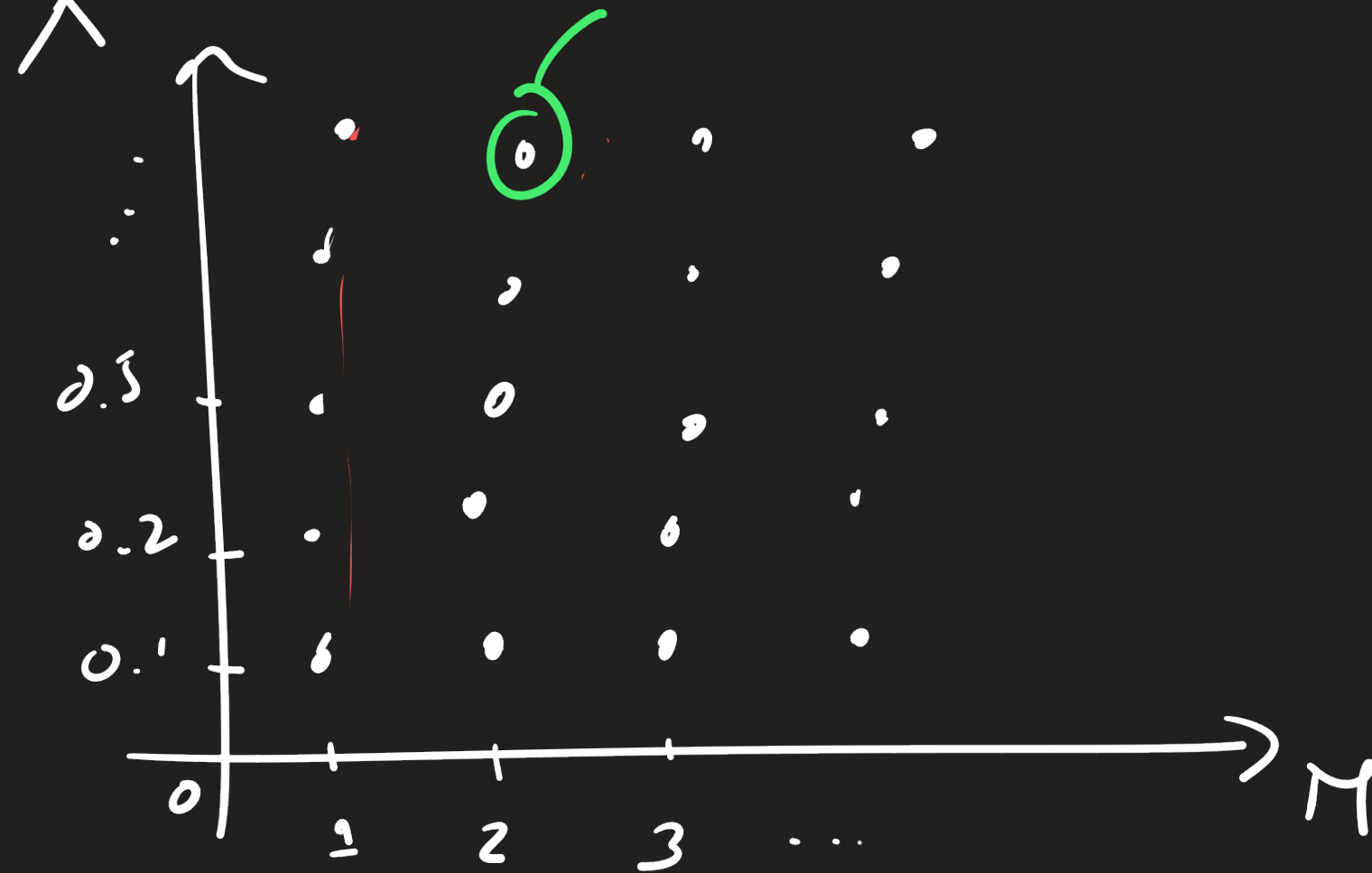
- 1) Compute the quantiles for each
and check if they match.





Cross-validation

tuning hyperparameters



Grid Search
OR
Factorial
Design.

Random
Search

for M in $[1, 2, 3, \dots]$:

for λ in $[0, 0.1, \dots]$:

K-FOLD CROSS-VALIDATION

illustrating with the
grid search approach.

TRAINING SET (80/20%)

S_1	S_2	S_3	S_4
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$K=4$
shuffle before
splitting into
evenly-sized
folds.

for M in $\{1, 2, 3, \dots\}$:

for λ in $\{0.01, 0.2, \dots\}$:

1) TRAIN SET: $\{S_1, S_2, S_3\} \rightarrow \vec{w}_1$
Validation: $\{S_4\}$

$$MSE_{train} = 1, MSE_{val} = 10$$

2) TRAIN SET: $\{S_1, S_2, S_4\} \rightarrow \vec{w}_2$
Validation: $\{S_3\}$

$$MSE_{train} = 100, MSE_{val} = 102$$

3) TRAIN: $\{S_1, S_3, S_4\} \rightarrow \vec{w}_3$
VAL: $\{S_2\}$

$$MSE_{train} = 0.01, MSE_{val} = 5$$

4) TRAIN: $\{S_2, S_3, S_4\} \rightarrow \vec{w}_4$
VAL: $\{S_1\}$

$$MSE_{train} = 5, MSE_{val} = 50$$

A) AVERAGE $MSE_{val} = \frac{10 + 102 + 5 + 50}{4}$

B) WORST performance in VAL = 102

C) STANDARD DEVIATION in VAL.

ONCE $L_{\text{test}}(M, \lambda)$ are found,

we train ^{the} ~~final~~ model using

the entire training set (80/20%)